GEYI YANG

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Research Interests

LLM-based agents, GUI agents, self-improving agentic systems, harness optimization, and retrieval-augmented generation (RAG).

Education

- **The Chinese University of Hong Kong, Shenzhen** *Sep. 2023 – Present*
  B.Eng. in Computer Engineering Shenzhen, China
  - GPA: 3.73/4.00.
  - Selected coursework: Artificial Intelligence, Machine Learning, Optimization, Data Structures, Operating Systems, and Internet Architecture and Protocol.
- **University of California, Berkeley** *Jan. 2026 – Jun. 2026*
  GLOBE Program Visiting Student in Computer Science Berkeley, CA, USA

Honors and Awards

COMAP Mathematical Contest in Modeling (MCM), Finalist (F Award; top 1%) 2026
China International College Students' Innovation Competition (CICSIC), Bronze Award 2025
Dean's List, CUHK-Shenzhen 2023–2024, 2024–2025, 2025–2026
Academic Performance Scholarship Awardee, CUHK-Shenzhen 2023–2024

Preprints

[1] **Geyi Yang**, Zikun Qu, Xiang Li, Zhiyong Wang, Min Zhang, Shipei Zeng, and Zhongxiang Dai. "GUI-HARVEST: Self-Improving GUI Agents through Evidence-Driven Harness Evolution." *arXiv preprint arXiv:2610.00948*, 2026. Under review at ICLR 2027. [arXiv]

Research Experience

- **GUI-HARVEST: Self-Improving GUI Agents through Evidence-Driven Harness Evolution** *Jul. 2026 – Present*
  Research Assistant, Shenzhen Research Institute of Big Data | *Advisor: Zhongxiang Dai* GitHub
  - First-authored GUI-HARVEST (under review at ICLR 2027), an automatic harness optimizer that enables self-improving GUI agents with frozen backbone models by converting repeated multimodal executions into validated runtime code changes.
  - Designed an evidence-driven loop that aligns actions with before-and-after screenshots, compares repeated runs, clusters recurring cross-task failures, and promotes edits only after score and predicted-behavior checks.
  - Achieved held-out gains across six open and proprietary backbones on OSWorld-Verified; Qwen3-VL-32B-Instruct gained 12.33 points on the full suite, while frozen-harness transfer improved GPT-5 by 13.87 points on WindowsAgentArena.
- **Meme-Qwen (7B-Instruct)** *Jan. 2026*
  Chinese Internet-Meme Conversational Language Model Model | Dataset
  - Independently built and open-sourced, to our knowledge, the first Chinese internet-meme conversational SFT dataset and the first open model of this type: collected 30,000+ post-comment pairs with Playwright and released 8,680 ChatML examples.
  - Fine-tuned Qwen2.5-7B on a single A100 (40GB) with LoRA and DPO; improved meme-use accuracy by 32%, with human and LLM preference rates of 82%/91%/68% for humor, accuracy, and naturalness.

Internships

- **Cloud Computing Center, Chinese Academy of Sciences** *Mar. 2025 – Jun. 2025*
  Domain-Specific Agentic RAG for Mining-Blasting QA Dongguan, China
  - Built and optimized an Agentic RAG system for retrieving and answering questions over mining-blasting technical documents, owning the end-to-end retrieval and reasoning pipeline.
  - Architected a hierarchical LangGraph workflow for memory, query planning, answer synthesis, and a ReAct subgraph for tool use and multi-step reasoning; added human clarification for ambiguous queries.
  - Developed Docling-based parsing and parent-child hybrid retrieval in Qdrant, combining BGE-M3, BM25, and BGE-Reranker-v2-m3; achieved 81% Recall@5 and 78% answer accuracy on 300+ QA cases.
- **Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)** *Sep. 2023 – Jun. 2024*
  Multimodal Digital Human Synthesis for Emotional Interaction Shenzhen, China
  - Developed an end-to-end "Digital Relative" pipeline from family photos and recordings, integrating MockingBird voice cloning, SadTalker lip-sync and facial animation, and ChatGPT-powered dialogue.
  - Piloted the system with family users in 500+ interactions: 83% reported a familiar emotional connection and 20% reported a strong emotional response.

Products

- **StashDock: Your Context Library for Any AI** *Oct. 2026 – Present*
  Creator and Full-Stack Developer Website | GitHub
  - Designed and launched an AI-native, visualized personal context library for managing and combining links, files, and prompts into reusable context sets. Enabled one-click addition to active AI conversations and AI-assisted reference discovery, filtering, and saving.
  - Engineered Windows/macOS desktop apps with Tauri/Rust, React/TypeScript, and SQLite, sharing one local library with a Chrome capture extension and a ChatGPT plugin that opens in the conversation sidebar.
  - Implemented visual reference selection, tagged context groups, floating-folder search, and clipboard/drag-and-drop transfer across AI apps; exposed reference search, content retrieval, tag management, and context saving through MCP tools and Skills.

Selected Projects

- **MCM 2026: Dynamic Hybrid Lunar Logistics Optimization** *2026*
  Modeling and Programming Lead Finalist (F Award; top 1%)
  - Developed a dynamic Earth-to-Moon logistics model integrating space elevators, reusable rockets, and ISRU maturation; formulated time-cost objectives and operational constraints.
  - Used Differential Evolution (1,301 evaluations), Monte Carlo analysis (5,000 simulations), and sensitivity analysis to identify a robust three-phase transition strategy; awarded Finalist (top 1%).
- **Telos: AI-Generated, Ready-to-Learn Courses** *Aug. 2026 – Present*
  Independent Full-Stack Developer GitHub
  - Building a goal-conditioned education product that turns user materials and cross-platform resources into structured, multimodal, ready-to-learn courses rather than isolated answers or learning plans.
  - Designed an agentic pipeline for retrieval, deduplication, quality assessment, sequencing, and embedding source text, images, and videos in a unified learning workspace.
- **Voice Turing Test: WeChat Mini Program and Research Data Platform** *May 2025 – Aug. 2025*
  Full-Stack Development Lead
  - Independently led registration, UI/UX, full-stack development, testing, and deployment of a gamified Human-vs-AI speech challenge used as the data-collection platform for *Human or Machine?* (ICLR 2026).
  - Built the native WeChat front end and Tencent Cloud backend with bilingual UI, real-time scoring, achievements, leaderboards, and referral rewards; reached 1,000+ users, 300+ DAU, and 5,000+ valid records in the first month.

Teaching Experience

- **The Chinese University of Hong Kong, Shenzhen** *Sep. 2025 – Dec. 2025*
  Teaching Assistant – MAT1001 Calculus I Shenzhen, China
  - Served as a teaching assistant for the first-year calculus course MAT1001, supporting tutorials and student learning.

Skills and Languages

Programming & Systems: Python, C++, JavaScript, TypeScript, Shell, FastAPI, React, SQLite, Git, Linux.
Machine Learning & Agents: PyTorch, Transformers, LoRA/PEFT, TRL (DPO), RAG, LangChain, LangGraph, GUI agents, Qdrant, MCP.
Languages: English (TOEFL 111; GRE 332).